

Uncertainty-Aware Bayesian Mineral Prospectivity Mapping under Spatial Separation: Heterogeneous Predictor Relationships across the Central African Copperbelt

Abstract

Whether geological and geophysical predictor relationships remain predictive across geographically separated portions of a mineralized belt, and how those relationships vary spatially, is a central question for regional mineral prospectivity mapping. We address this question in the Central African Copperbelt using a partially pooled hierarchical Bayesian logistic model evaluated across four geographically contiguous held-out sectors. The 1,872-cell modelling frame (5 km $\times$ 5 km grid; 138 documented sediment-hosted stratiform copper-cobalt occurrences) includes distance to mapped faults, distance to lithological contacts, Bouguer gravity, and retained host-lithology classes as predictors. Intercepts and the four distance coefficients (linear and quadratic, constructed from standardized distances) vary by Daly tectonic domain; Bouguer gravity and host-lithology effects are global. Posterior draws provide uncertainty distributions for cell-level conditional occurrence probabilities $p_i^{(s)} = \text{logit}^{-1}(\eta_i^{(s)})$, while bootstrap intervals quantify uncertainty in held-out performance metrics. Predictor contributions to the out-of-fold linear predictor shift markedly across along-belt sectors: fault-distance dominance decreases from north to south as Bouguer gravity becomes co-dominant. Across four distance-representation specifications, mean ROC-AUC ranges from 0.671 to 0.686 and mean PR-AUC from 0.107 to 0.145; distance associations recur, but quadratic curvature is representation-dependent and the population-level stationary-point diagnostic D^* is not identifiable. The results establish spatially heterogeneous fitted predictor relationships and heterogeneous geographic predictive transferability, but do not establish causal geological mechanisms or a universal distance relationship.

Keywords: mineral prospectivity mapping; Bayesian hierarchical model; spatial cross-validation; spatial heterogeneity; Central African Copperbelt; Daly tectonic domains; parameter uncertainty

1 Introduction

Mineral prospectivity mapping integrates geological, geochemical, and geophysical evidence to identify locations whose mapped characteristics are associated with known deposits. Published approaches range from knowledge-driven evidence integration (Bonham-Carter, 1994; Carranza, 2009) to logistic regression, weights of evidence (Bonham-Carter et al., 1989), Bayesian networks, and machine-learning classifiers (Porwal et al., 2006; Carranza and Laborte, 2015; Ford, 2019; Yousefi et al., 2021; Matthieu et al., 2026); they differ in assumptions, data requirements, and treatment of incomplete deposit inventories. In the Central African Copperbelt, stratigraphy, structures, basin evolution, and ore assemblages vary regionally, while the timing and detailed genesis of mineralization remain debated (Cailteux et al., 2005; Selley et al., 2005; Hitzman et al., 2012; Key et al.,

2004; Ndonfack et al., 2025). The predictors used here are spatial proxies within that geological system, not direct measurements of ore-forming processes.

Spatial structure also affects evaluation. When nearby observations share covariates, occurrence history, or sampling processes, random cross-validation can place dependent observations in both training and test sets and underestimate error for a new-region prediction task (Brenning, 2012; Roberts et al., 2017). The size and direction of this bias depend on the prediction target and blocking design; spatial blocks can also create predictor extrapolation (Valavi et al., 2019). This study therefore evaluates transfer among four contiguous along-belt sectors rather than treating random cell splitting as a proxy for geographic transfer.

The Bayesian formulation represents posterior uncertainty in shared parameters, hierarchical scales, and domain effects. Draw-wise held-out conditional probabilities are averaged to produce stored OOF scores, but the frozen OOF artifact does not retain cell-level probability intervals. Future Bernoulli outcomes are not sampled. Uncertainty in evidence layers, model parameters, conditional probabilities, and outcomes are distinct quantities in prospectivity work (Lisitsin et al., 2014; Parsa and Carranza, 2021; Zuo et al., 2021).

The Central African Copperbelt provides a setting in which to examine regional transfer. Daly’s domain labels are used as a geological stratification for model terms and secondary summaries (Daly et al., 2025); the model estimates associations indexed by those labels but does not validate the tectonic framework itself. Spatially varying coefficient models offer a statistical representation of changing relationships without making that variation causal (Gelfand et al., 2003; Fotheringham et al., 2002).

This study contributes: (i) a partially pooled hierarchical Bayesian mineral prospectivity framework evaluated under along-belt spatial separation to assess geographic transferability while reducing local train-test overlap; and (ii) a representation-sensitivity design that separates recurring proximity associations from representation-dependent polynomial curvature and D^* stationary-point diagnostics.

The objective is to estimate multivariate geological and geophysical predictor associations under a hierarchical Bayesian model and evaluate how fitted relationships and discrimination vary across four spatially separated Copperbelt sectors. We ask how fault distance, lithological-contact distance, Bouguer gravity, and host lithology jointly relate to the recorded occurrence labels; how model-component contributions differ across held-out sectors; how OOF discrimination varies; and whether distance-response conclusions persist under alternative mathematical representations.

2 Study area, spatial data, and evidence inventory

2.1 Study area and grid definition

The study area encompasses the Central African Copperbelt tract represented in the project spatial modelling grid, spanning the border region of northern Zambia and the southern Democratic Republic of the Congo. The spatial frame is defined on a regular $5\text{ km} \times 5\text{ km}$ grid projected in WGS 84 / UTM Zone 35S (EPSG:32735). The grid extends from Easting 282,675 m to 715,465 m and Northing 8,380,251 m to 8,859,512 m, covering an along-belt tract of approximately 500 km in strike length.

The observational unit is one 5 km by 5 km grid cell identified by its centroid coordinates (x_i, y_i)

and a binary occurrence indicator $Y_i \in \{0, 1\}$. The supplied grid is in WGS 84 / UTM Zone 35S (EPSG:32735), the projected coordinate system used for stored centroids and distance fields. The V11 code excludes rows with a missing centroid coordinate, raw domain label, host-lithology class, fault distance, lithology-contact distance, or Bouguer value; it then excludes domain labels that do not map to one of its six recognized categories. `deposit_present` is not included in the code’s `dropna` list; the supplied modeling table contains labels for retained rows. These filters yield 1,872 cells, including 138 positive and 1,734 background cells.

The target is presence, not an occurrence count. The available base-grid table contains 210 positive rows across 139 unique cell IDs (71 repeated positive rows). The executed feature notebook deduplicates the base target by `id`, retaining the first row, before merging the binary flag onto unique cells; repeated rows therefore do not become repeated V11 observations. The original occurrence inventory is not present, so the repeated rows cannot be classified further as distinct deposits, overlapping footprints, or duplicate records.

2.2 Data sources and feature extraction

V11 reads the prepared fields from `data/copperbelt_training_v5_with_tectonic_domain.csv`; the fit does not reproject source layers or calculate raw GIS distances. The supplied project layers and executed feature notebook support the following provenance statements:

1. **Distance to mapped faults (d_f):** The supplied `CACB_Structures.gpkg` contains a `faults_shear_zones` layer whose `source` attribute identifies `Selley_et_al.2005.Fig1` (Selley et al., 2005). V11 consumes the precomputed `distance_to_fault` field in projected meters.
2. **Distance to lithological contacts (d_l):** The prepared contact-distance and `all_lithology_lines.gpkg` layers provide the precomputed `distance_to_lithology_contact` and `litho_contact_litho_class` fields. The inspected layer attributes do not identify the original geological map or its scale; that original provenance cannot be established from the available files. V11 consumes the stored distance in projected meters.
3. **Bouguer gravity (g):** An executed project notebook reads a regional text grid named `grid_bouguer_-17.558.5` with longitude, latitude, and Bouguer columns. It transforms the grid coordinates from EPSG:4326 to EPSG:32735 and assigns each cell the value of the nearest grid point using a $k = 1$ KD-tree query. The source text grid and original publisher metadata are absent from the repository and supplied external data directory; a WGM2012/BGI attribution could not be verified.
4. **Lithology class ($\mathbf{x}_{\text{rock}}$):** The V11 categorical predictor is `litho_contact_litho_class`, associated with the supplied project contact-linework data. It is not the separate `Africa_Surface.Lithology.tif` raster. Its original geological-map provenance is not documented in the inspected files.
5. **Tectonic domain stratification ($d(i)$):** V11 reads the `domain` field from the input table and maps its string values to the six labels NRB.3a, NRB.3b, CRZ, NKB, SRB, and MMSB using substring rules. The V11 fit does not digitize polygons or perform point-in-polygon assignment. Daly’s framework provides geological context for these labels (Daly et al., 2025), but the provenance of the input table’s cell-level domain assignments is not documented in the supplied project files.

2.3 Domain-level positive-observation structure and partial-pooling qualification

Documented deposit occurrences in the modelling frame are concentrated exclusively in two of the six tectonic domains: 107 positive cells occur in NRB_3a (out of 1,127 total cells) and 31 occur in NRB_3b (out of 550 total cells). The remaining four domains contain zero recorded positive observations in this compiled frame: CRZ ($n = 154$), NKB ($n = 24$), MMSB ($n = 16$), and SRB ($n = 1$).

Methodologically, this empirical structure has specific implications for the hierarchical Bayesian model:

- The formulation remains a fully partially pooled hierarchical model across all six domains, with domain-varying parameters linked through shared population hyperparameters μ_p and σ_p .
- In CRZ, NKB, SRB, and MMSB, the likelihood contains no within-domain positive-class information. Their domain-specific effects are informed by the available negative observations together with the shared hierarchical population structure, and therefore rely more strongly on partial pooling than effects in domains containing both classes.
- Consequently, domain-specific coefficient interpretations for CRZ, NKB, SRB, and MMSB must be treated with substantial caution, and standard binary classification metrics such as ROC-AUC are undefined within these single-class partitions.

2.4 Complete preprocessing chain

The complete preprocessing and spatial prediction workflow is shown in Figure 1. V11 removes rows missing either centroid coordinate, `domain`, `litho_contact_litho_class`, `distance_to_fault`, `distance_to_litho`, or `bouguer`, then excludes unrecognized domain labels. Within each fold, standardization statistics and lithology support are determined from training cells only and applied to held-out cells; this prevents leakage through those preprocessing operations. Each continuous predictor is standardized as $z_i = (X_i - \mu_{\text{train}})/\sigma_{\text{train}}$. Quadratic distance terms are constructed as $z_{f,i}^2$ and $z_{l,i}^2$ after standardization. A one-hot lithology category is retained only if the training partition contains at least one positive and at least one negative cell in that category; test-fold categories do not affect this filter.

2.5 Distinction between validation folds and tectonic domains

It is essential to distinguish the two spatial stratifications used in this study:

- **Along-belt validation folds (Folds 1–4):** Centroid coordinates are subjected to PCA; observations are projected onto the first principal component, sorted by that projection, and divided into four approximately equal contiguous groups of 468 cells. Each group is held out once to evaluate geographic transfer. This reduces local train-test overlap but does not guarantee independence across fold boundaries.
- **Daly tectonic domains (6 domains):** A geological and structural stratification based on Katangan basin evolution (Daly et al., 2025), used as the grouping variable for hierarchical partial pooling of model parameters.

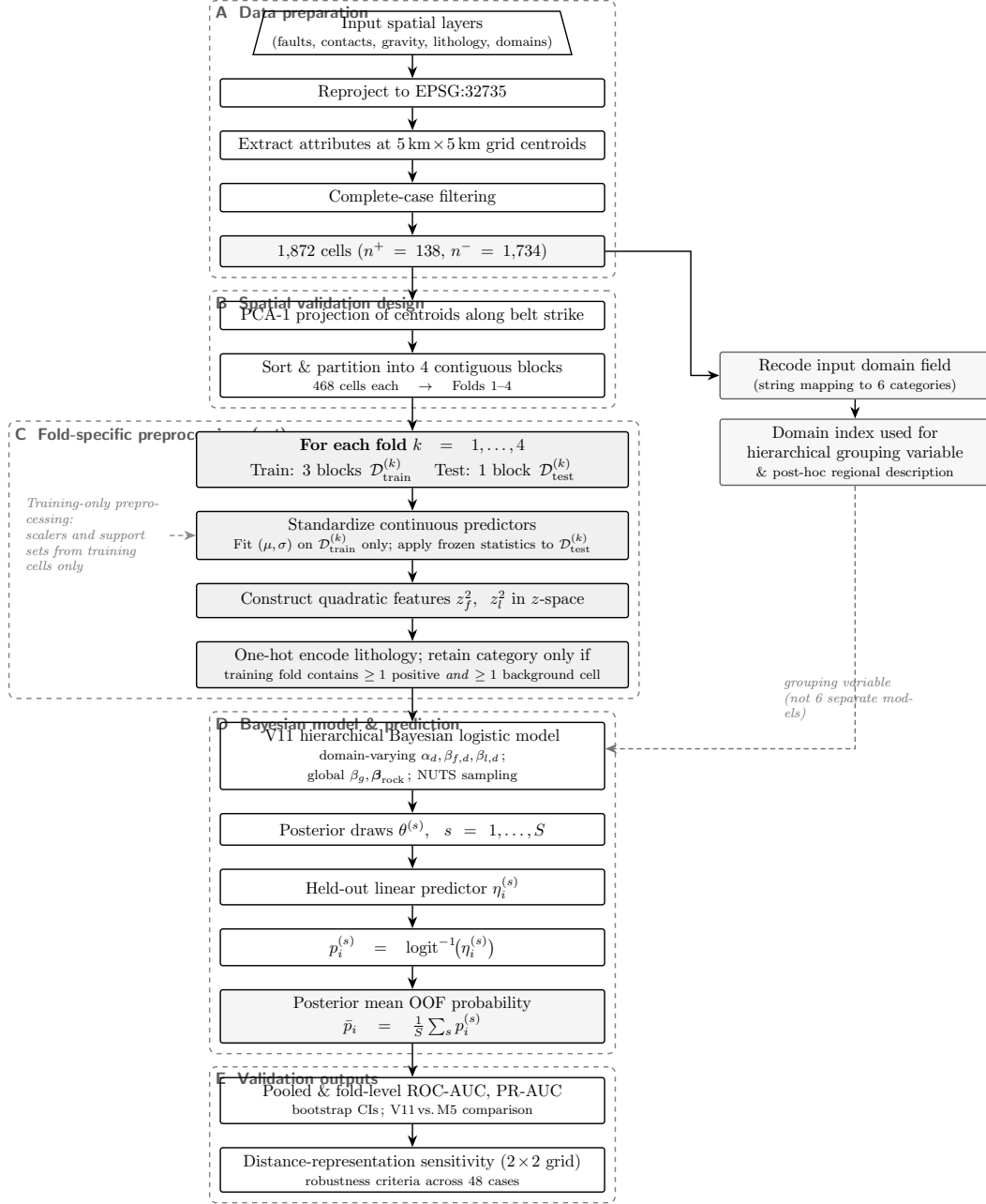


Figure 1: Complete preprocessing and spatial prediction workflow for the V11 hierarchical Bayesian prospectivity model. Centroids are projected onto the first principal component, sorted, and divided into four approximately equal along-belt folds. Fold-specific standardization and host-lithology support filtering use training cells only; held-out cells use those frozen training-fold operations.

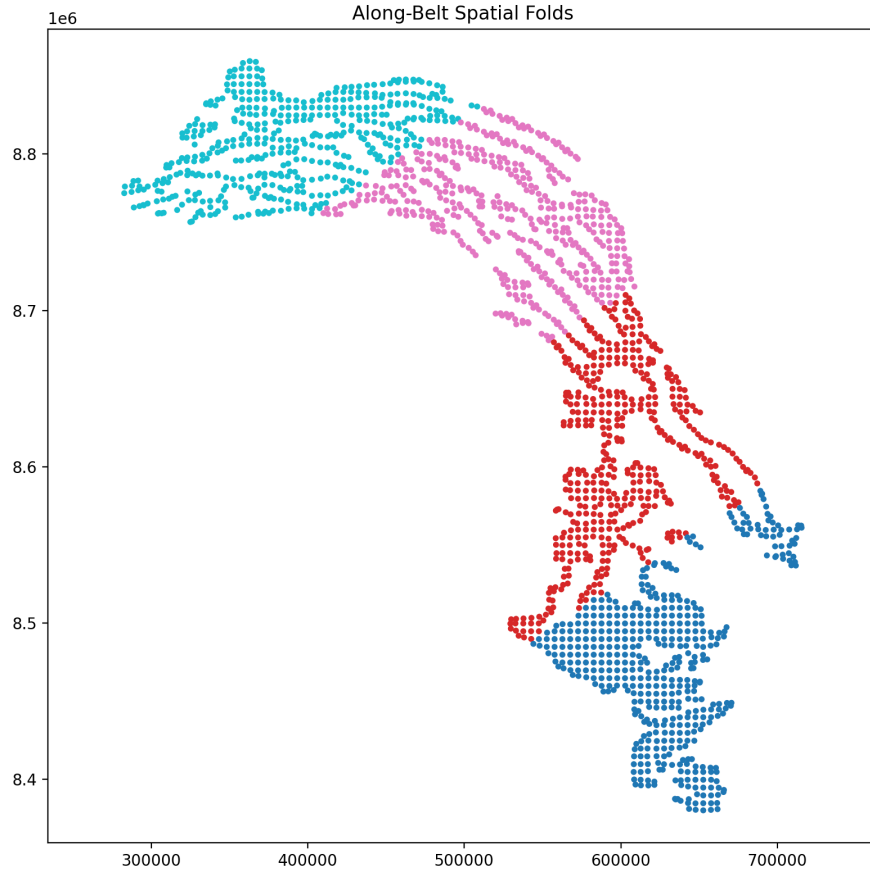


Figure 2: Four along-belt spatial validation folds across the Central African Copperbelt modeling grid. Centroid coordinates are projected onto their first principal component, ordered by projection, and divided into four balanced folds of 468 cells each; one fold is held out at a time.

The validation is neither Leave-One-Daly-Domain-Out nor a Fold $\times$ Domain cross-validation. The model is trained on three spatial folds and evaluated on the held-out spatial fold; domain-level performance metrics are computed post hoc as descriptive strata of the spatial predictions.

3 Bayesian hierarchical methodology and spatial validation

3.1 V11 model specification

For cell i in Daly domain $d(i) \in \{1, \dots, 6\}$, the V11 model is a partially pooled hierarchical Bayesian logistic regression (Gelman and Hill, 2007):

$$Y_i \sim \text{Bernoulli}(p_i), \quad \text{logit}(p_i) = \eta_i, \quad (1)$$

where the linear predictor η_i decomposes as:

$$\eta_i = \alpha_{d(i)} + \beta_{f,\text{lin},d(i)} z_{f,i} + \beta_{f,\text{sq},d(i)} z_{f,i}^2 + \beta_{l,\text{lin},d(i)} z_{l,i} + \beta_{l,\text{sq},d(i)} z_{l,i}^2 + \beta_g z_{g,i} + \mathbf{x}_{\text{rock},i}^\top \boldsymbol{\beta}_{\text{rock}}. \quad (2)$$

The model distinguishes:

- **Domain-varying parameters:** Domain intercepts α_d and the four distance coefficients (fault linear $\beta_{f,\text{lin},d}$, fault quadratic $\beta_{f,\text{sq},d}$, lithological contact linear $\beta_{l,\text{lin},d}$, and lithological contact quadratic $\beta_{l,\text{sq},d}$).
- **Global parameters:** The Bouguer gravity coefficient β_g and the host-lithology vector $\boldsymbol{\beta}_{\text{rock}}$ are shared globally across all domains.

Domain-varying parameters use a non-centred hierarchical parameterization to ensure efficient MCMC exploration:

$$\beta_{p,d} = \mu_p + \sigma_p \cdot \text{offset}_{p,d}, \quad \text{for } p \in \{f_{\text{lin}}, f_{\text{sq}}, l_{\text{lin}}, l_{\text{sq}}\}, \quad (3)$$

and

$$\alpha_d = \mu_\alpha + \sigma_\alpha \cdot \text{offset}_{\alpha,d}. \quad (4)$$

The model uses the following priors:

$$\mu_p \sim \mathcal{N}(0, 1), \quad \sigma_p \sim \text{HalfNormal}(1), \quad (5)$$

$$\text{offset}_{p,d} \sim \mathcal{N}(0, 1), \quad \mu_\alpha \sim \mathcal{N}(\text{logit}(\hat{p}_{\text{train}}), 1), \quad (6)$$

$$\sigma_\alpha \sim \text{HalfNormal}(1), \quad \text{offset}_{\alpha,d} \sim \mathcal{N}(0, 1), \quad (7)$$

$$\beta_g \sim \mathcal{N}(0, 1), \quad \boldsymbol{\beta}_{\text{rock}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (8)$$

where $\hat{p}_{\text{train}} = \sum y_{\text{train}} / N_{\text{train}}$ anchors the intercept prior to the empirical training base rate.

3.2 Continuous probability prediction and threshold-free evaluation

For each posterior draw $s \in \{1, \dots, S\}$ and held-out cell i , the prediction code computes the draw-specific linear predictor from that draw’s parameters and the held-out cell’s fold-preprocessed features, then applies the logistic link to obtain its conditional prospectivity probability:

$$\eta_i^{(s)} = \alpha_{d(i)}^{(s)} + \dots, \quad p_i^{(s)} = \text{logit}^{-1}(\eta_i^{(s)}) = \frac{1}{1 + \exp(-\eta_i^{(s)})}. \quad (9)$$

The reported out-of-fold prediction is the posterior mean conditional probability:

$$\bar{p}_i = \frac{1}{S} \sum_{s=1}^S p_i^{(s)}. \quad (10)$$

No classification threshold (e.g., 0.5) was applied to generate the reported OOF predictions or calculate ROC-AUC/PR-AUC. Predictive performance is evaluated across the continuous distribution of predicted probabilities using threshold-independent rank discrimination metrics (Lobo et al., 2008).

3.3 Uncertainty concepts

We maintain four distinct uncertainty concepts:

1. **Parameter posterior uncertainty:** Represented by MCMC posterior samples of coefficients and hyperparameters.
2. **Posterior uncertainty in the conditional probability:** For each held-out cell and draw s , the prediction logic computes $\eta_i^{(s)}$ and $p_i^{(s)} = \text{logit}^{-1}(\eta_i^{(s)})$. The distribution $\{p_i^{(s)}\}_{s=1}^{3000}$ is posterior uncertainty in the cell’s conditional probability induced by parameter uncertainty; it is not a Bernoulli outcome interval. The frozen OOF CSV stores only $\bar{p}_i = 3000^{-1} \sum_s p_i^{(s)}$. A post-fit reconstruction from the four frozen traces reproduced all 1,872 frozen OOF means, with maximum absolute difference approximately 9.97×10^{-17} , without fitting a model. Cell-level posterior summaries were not in the original frozen prediction table.
3. **Future outcome uncertainty:** The workflow does not sample binary Bernoulli realizations $Y_i^{(s)} \sim \text{Bernoulli}(p_i^{(s)})$; predictions represent expected conditional probabilities, not simulated futures.
4. **Performance metric uncertainty:** The final fold and pooled metric intervals use paired row-bootstrap resampling of fixed OOF scores. A separate multiscale spatial block bootstrap resamples occupied coordinate blocks and evaluates robustness of ΔAUC at the specified block scales. Both quantify uncertainty in performance metrics; neither is cell-level posterior probability uncertainty.

3.4 Decomposition of linear predictor variance

To describe how additive predictor components vary across held-out cells in each sector, the out-of-fold linear predictor is decomposed for each test cell i in fold k :

$$\eta_i = c_{\text{intercept},i} + c_{\text{fault},i} + c_{\text{lith},i} + c_{\text{grav},i} + c_{\text{rocks},i}, \quad (11)$$

where each component represents the posterior expectation across retained MCMC draws:

$$c_{\text{fault},i} = \mathbb{E}_{\text{post}} [\beta_{f,\text{lin},d(i)} z_{f,i} + \beta_{f,\text{sq},d(i)} z_{f,i}^2], \quad (12)$$

$$c_{\text{lith},i} = \mathbb{E}_{\text{post}} [\beta_{l,\text{lin},d(i)} z_{l,i} + \beta_{l,\text{sq},d(i)} z_{l,i}^2], \quad (13)$$

$$c_{\text{grav},i} = \mathbb{E}_{\text{post}} [\beta_g z_{g,i}], \quad (14)$$

$$c_{\text{rocks},i} = \mathbb{E}_{\text{post}} [\mathbf{x}_{\text{rock},i}^\top \boldsymbol{\beta}_{\text{rock}}]. \quad (15)$$

For each predictor component $m \in \{\text{fault}, \text{lith}, \text{grav}, \text{rocks}\}$, its dispersion across the N_{test} held-out cells of fold k is measured by its sample standard deviation:

$$\text{SD}_m = \sqrt{\frac{1}{N_{\text{test}}} \sum_{i \in \mathcal{D}_{\text{test}}^{(k)}} (c_{m,i} - \bar{c}_m)^2}. \quad (16)$$

The reported component share is the component standard deviation divided by the sum of the four component standard deviations. It is therefore an SD-normalized share of additive-component dispersion, not a proportion of outcome variance explained:

$$\text{Share}_m = \frac{\text{SD}_m}{\text{SD}_{\text{fault}} + \text{SD}_{\text{lith}} + \text{SD}_{\text{grav}} + \text{SD}_{\text{rocks}}} \times 100\%. \quad (17)$$

These values describe relative dispersion of the additive components of the decomposed OOF linear predictor across test cells. They are not causal geological effects, Shapley values, or proportions of outcome variance explained (R^2). Because the implementation normalizes component standard deviations (rather than squared variances), the percentages should be read as SD-normalized contribution shares.

3.5 Distance-representation sensitivity design and robustness criteria

To test whether distance-response inference depends on mathematical representation, four models form a 2×2 factorial grid:

Table 1: Distance-representation sensitivity design (2×2 grid).

	Quadratic	Linear
Raw distance (z)	Model A (V11)	Model C
Log distance ($\log(1 + x_{\text{km}})$)	Model B	Model D

Models A and C use standardized raw distance; Models B and D use standardized $\log(1 + x_{\text{km}})$. The compact M5 baseline is a non-hierarchical logistic model containing only standardized Bouguer gravity and lithological-contact distance with its quadratic term.

Across the four spatial folds, six Daly domains, and two distance predictors (fault and lithological contact), there are $4 \times 6 \times 2 = 48$ evaluable comparison cases. In each case, Model A (raw-quadratic) is compared against Model B (log-quadratic) using three pre-registered criteria:

1. **Criterion 1 (Curvature probability agreement):** The absolute difference in posterior probability of positive quadratic curvature between representations must be less than 0.15:

$$|P(\beta_{\text{sq}} > 0 \mid \text{Model A}) - P(\beta_{\text{sq}} > 0 \mid \text{Model B})| < 0.15. \quad (18)$$

2. **Criterion 2 (D^* median agreement):** The relative difference between posterior median stationary points D_A^* and D_B^* (in physical kilometers) must not exceed 25%:

$$\frac{|D_A^* - D_B^*|}{\max(|D_A^*|, |D_B^*|)} \leq 0.25. \quad (19)$$

3. **Criterion 3 (Data support concentration):** Both median stationary points must fall within their respective training data empirical support ranges:

$$\text{support_min}_A \leq D_A^* \leq \text{support_max}_A \quad \text{and} \quad \text{support_min}_B \leq D_B^* \leq \text{support_max}_B. \quad (20)$$

A relationship is classified as representationally robust if and only if it satisfies all three criteria simultaneously.

4 Results

4.1 Multivariate posterior structure

The fitted V11 model jointly estimates fault distance, lithological-contact distance, Bouguer gravity, and host lithology across domains and folds. In NRB_3a, lithological-contact distance has negative linear coefficient medians (-0.849 to -0.451) and positive quadratic coefficient medians ($+0.293$ to $+0.334$), with posterior probabilities of positive curvature $P(\beta_{l,\text{sq}} > 0) \geq 0.993$ across all four folds (Figure 4). In NRB_3b, posterior support for positive curvature is fold-dependent and weaker than in NRB_3a.

Fault-distance effects vary more dramatically (Figure 3). In NRB_3a, linear medians range from -2.681 to -0.703 , while quadratic medians are negative in Fold 2 and positive in Folds 1, 3, and 4. In NRB_3b, fault linear medians shift to positive values ($+0.181$ to $+1.548$), demonstrating that the directional association with fault distance is not spatially uniform across the belt.

4.2 Spatial heterogeneity in predictor contributions

The SD-normalized shares of additive out-of-fold linear-predictor components vary across along-belt sectors (Table 2 and Figure 5):

In Fold 1, the fault-distance component has the largest SD-normalized share (86.5%). Its share is 50.3% in Fold 2, 33.1% in Fold 3, and 34.9% in Fold 4. The Bouguer component share ranges from 2.0% in Fold 1 to 31.1% in Fold 4. Host lithology and contact-distance component shares are largest in Folds 2 and 3. These percentages are descriptive features of the decomposed OOF linear predictor and do not establish geological controls.

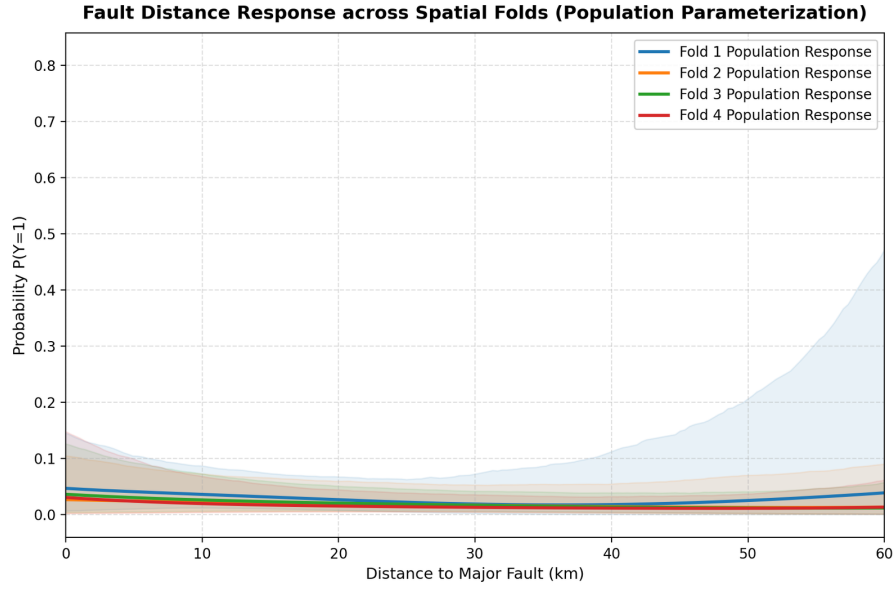


Figure 3: Fold-specific posterior response curves for fault distance across Daly domains under the V11 model. Shaded bands represent posterior credible intervals, illustrating substantial variation across along-belt sectors.

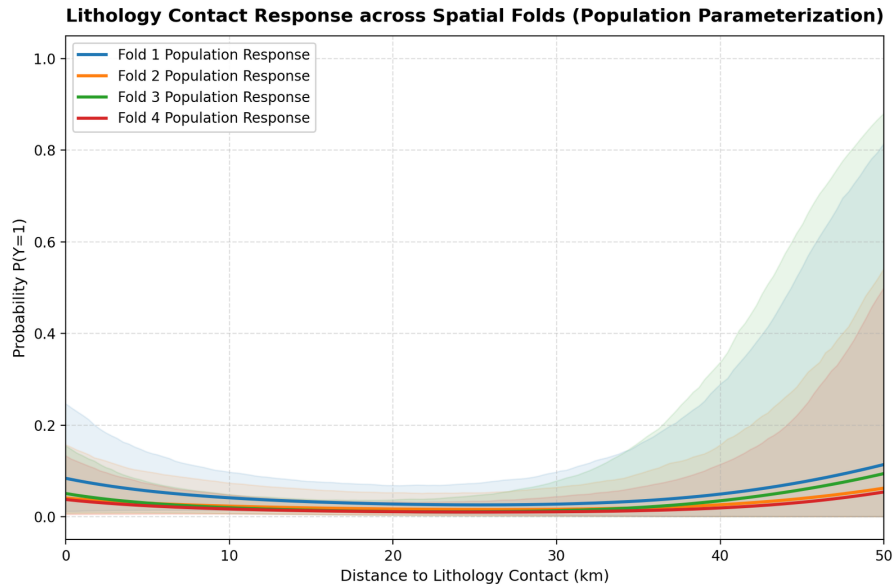


Figure 4: Fold-specific posterior response curves for lithological-contact distance across Daly domains under the V11 model. NRB_3a exhibits recurring positive quadratic curvature, whereas NRB_3b shows broader dispersion.

Table 2: SD-normalized shares of decomposed out-of-fold linear-predictor components.

Fold	Fault distance	Lithology distance	Bouguer gravity	Host lithology
1	86.5%	11.1%	2.0%	0.4%
2	50.3%	23.9%	5.6%	20.3%
3	33.1%	28.2%	15.3%	23.4%
4	34.9%	17.7%	31.1%	16.3%

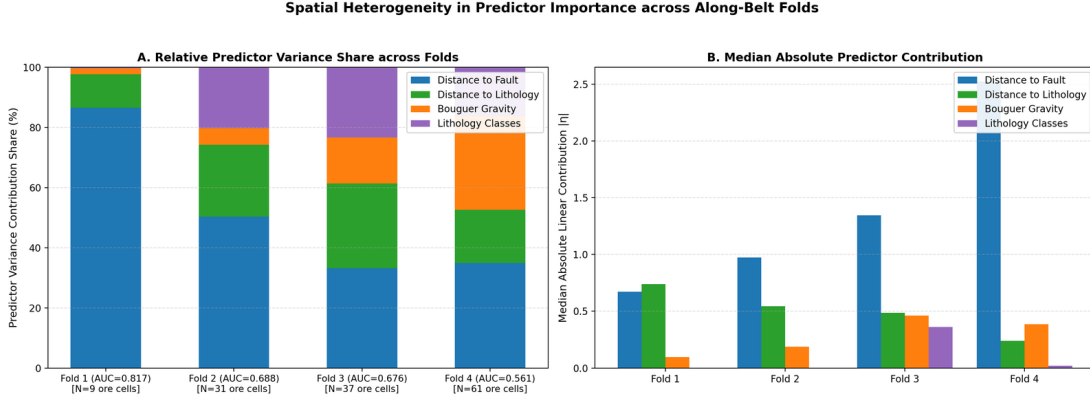


Figure 5: SD-normalized contribution shares of additive components in the decomposed out-of-fold linear predictor across the four spatial folds. Shares describe model-predictor dispersion, not causal geological importance.

4.3 Distance-response sensitivity and diagnostic failure of D^*

Mean spatial discrimination was similar across the four evaluated representations: mean ROC-AUC values are 0.686 for Model A, 0.673 for Model B, 0.671 for Model C, and 0.683 for Model D. Mean PR-AUC values are 0.107, 0.123, 0.130, and 0.145, with the log-linear specification (Model D) achieving the highest PR-AUC. Similar fold-averaged scores do not establish that the underlying response shapes are invariant.

While proximity associations persist, polynomial curvature and the derived stationary point $D^* = -\beta_{\text{lin}}/(2\beta_{\text{sq}})$ are representation-dependent. Pre-registered robustness criteria across the 48 evaluation cases passed in 13 of 48 cases for curvature sign agreement (Criterion 1), 0 of 48 cases for stationary-point agreement (Criterion 2), and 15 of 48 cases for posterior support concentration (Criterion 3). All three criteria were satisfied simultaneously in **0 of 48 cases**.

At the population level, posterior curvature distributions overlap zero: $P(\mu_{f,\text{sq}} > 0) = 0.384$ and $P(\mu_{l,\text{sq}} > 0) = 0.743$. The 95% credible intervals for population D^* span $[-279.36, 366.38]$ km for fault distance and $[-161.69, 152.34]$ km for lithological contact distance. These unphysically wide intervals include negative distances and values far exceeding the basin width, demonstrating that population-level D^* is statistically unidentifiable and cannot be interpreted as a physical trap distance.

4.4 Spatial predictive transferability and prospectivity mapping

The out-of-fold prospectivity predictions produce the regional prospectivity map shown in Figure 6. Across the four spatial folds, V11 achieves fold-level ROC-AUC values of 0.817, 0.688, 0.676, and 0.561, with PR-AUC values of 0.048, 0.103, 0.133, and 0.145. Pooled across all 1,872 cells, V11 achieves ROC-AUC of 0.689 [95% bootstrap CI: 0.647, 0.727], compared with 0.525 [0.470, 0.581] for the M5 baseline, yielding a pooled gain of +0.164 [+0.084, +0.236].

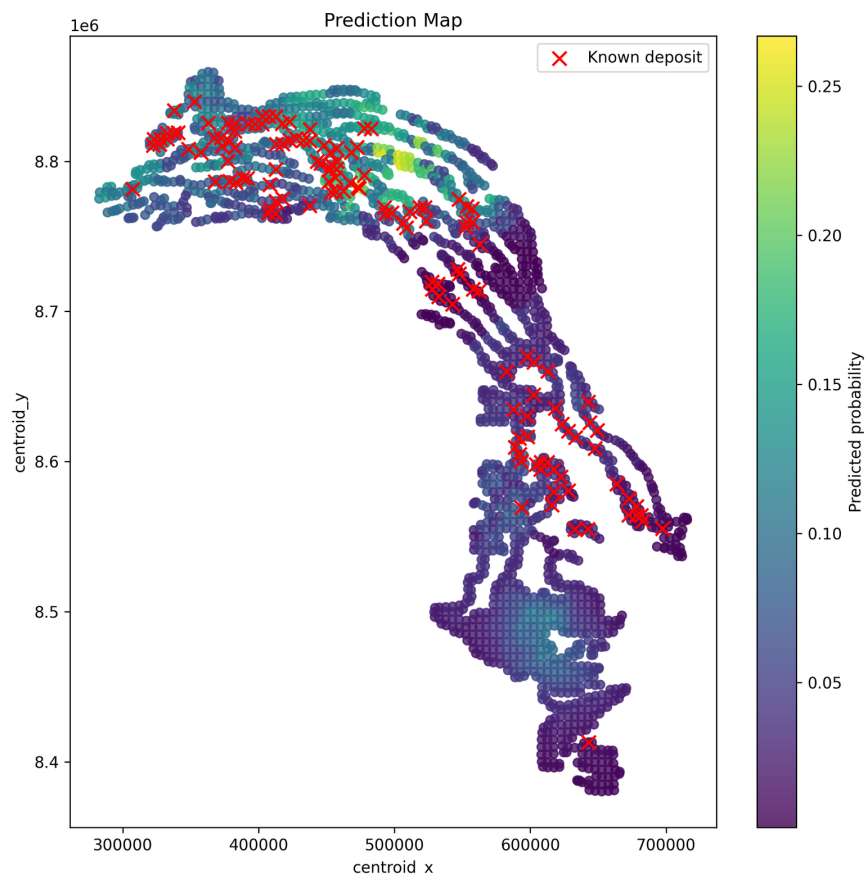


Figure 6: Out-of-fold Bayesian conditional prospectivity map ($\bar{p}_i$) across the Central African Copperbelt under the V11 model. Continuous predicted probabilities reflect spatial variation in prospective potential under along-belt hold-out evaluation.

Pooled performance masks geographic divergence: in Fold 3, V11 exceeds M5 by $\Delta\text{AUC} = +0.091$ (95% paired row-bootstrap CI [+0.024, +0.159]); in Fold 2, V11 trails M5 by $\Delta\text{AUC} = -0.136$ (95% CI [-0.250, -0.020]). The corresponding intervals for Folds 1 and 4 span zero. These differences describe performance on the specified held-out folds and do not identify their geological cause.

In secondary domain stratification, NRB_3a V11 ROC-AUC is 0.661701 versus M5 0.535752 ($\Delta = +0.125948$). In NRB_3b, V11 ROC-AUC is 0.583815 versus M5 0.866897 ($\Delta = -0.283082$). ROC-AUC is undefined in CRZ, NKB, SRB, and MMSB because these strata contain no positive observations.

5 Interpretation and representation sensitivity

5.1 Conditional multivariate inference and predictor association

The multivariate results establish that geological and geophysical predictors contain joint predictive signal in the evaluated Copperbelt modelling frame. Fault distance, lithological-contact distance, Bouguer gravity, and host lithology contribute jointly to the fitted out-of-fold predictor. The hierarchical structure permits domain-varying intercepts and distance effects while sharing information across Daly domains. This conditional structure means that individual coefficients reflect predictive associations holding other variables constant, rather than independent marginal relationships.

5.2 Spatial non-stationarity across along-belt sectors

The OOF decomposition shows that the SD-normalized shares of additive predictor components vary across along-belt sectors. The fault-distance share is largest in Fold 1; contact distance and host lithology have larger shares in Folds 2 and 3; and gravity’s share rises in Fold 4. These are model-based descriptive patterns rather than evidence of causal geological controls. The regional geological literature documents variation in basin structure and stratigraphic setting, providing context for examining these patterns without identifying their cause (Cailteux et al., 2005; Selley et al., 2005; Hitzman et al., 2012; Ndonfack et al., 2025).

5.3 Representation sensitivity and diagnostic role of D^*

Proximity associations persist across mathematical transformations, but polynomial curvature and the derived stationary point D^* are sensitive to functional form. The zero-pass rate across the 48 robustness cases demonstrates that D^* is a model-dependent algebraic property of quadratic curves rather than an invariant physical optimum. Consequently, D^* should be treated as an audit diagnostic for polynomial parameterizations, and prospectivity mapping should avoid translating quadratic extrema into universal exploration drilling targets.

6 Discussion, limitations, and exploration implications

6.1 What the model establishes: empirical association versus causal ore genesis

The results show that geological proxies contain predictive signal for sediment-hosted copper prospectivity, with fitted associations and held-out discrimination varying across the evaluated sectors. The SD-normalized shares describe dispersion of additive components in the posterior-mean OOF linear predictor; they are neither causal effects nor a direct measure of each variable’s marginal or conditional predictive value. Bouguer gravity’s larger share in Fold 4 is descriptive and does not establish a geological mechanism or explain regional differences in discrimination.

6.2 Regional transferability and Daly’s tectonic framework

Spatially separated prediction demonstrates that pooled belt-wide skill can conceal regional performance divergence. V11 outperforms the baseline in Fold 3 but underperforms it in Fold 2. This

divergence demonstrates that prospectivity models cannot be assumed to transfer uniformly across a mineralized belt, reinforcing the necessity of spatially separated validation strategies (Roberts et al., 2017; Valavi et al., 2019).

Daly’s tectonic domain classification (Daly et al., 2025) provides a principled geological stratification for hierarchical partial pooling. While the model accommodates domain-level parameter differences, the analysis estimates empirical associations within the compiled data and does not constitute a formal proof of Daly’s tectonic hypothesis in its entirety.

6.3 Model parsimony and distance representations

The comparable discrimination achieved by linear and log-linear specifications (Models C and D) relative to quadratic models indicates that quadratic distance terms did not improve mean discrimination in these evaluated folds. Linear and logarithmic representations are simpler specifications for this comparison; the results do not establish a universally preferred distance form.

6.4 Methodological and data limitations

Several methodological limitations must be acknowledged:

1. **Sample size and domain imbalance:** Only 138 deposit-present cells are available, with 107 concentrated in NRB_3a and 31 in NRB_3b. Four Daly domains contain zero positive observations in this modelling frame, precluding within-domain discrimination evaluation. Their domain effects are still estimated, but lack within-domain positive-class information and depend more strongly on the shared hierarchical structure.
2. **Residual spatial autocorrelation:** Although contiguous spatial blocking mitigates optimistic cross-validation bias, spatial autocorrelation may persist across block boundaries (Roberts et al., 2017).
3. **Input layer uncertainty:** Regional geological maps, structural compilations, and gravity grids contain measurement, interpolation, and cartographic errors that propagate into distance calculations.
4. **Surface proxies vs. 3D geometry:** Near-surface distance features do not capture three-dimensional fault dip, subsurface stratigraphy, or depth to basement.
5. **Discrete blocking approximation:** Four discrete spatial folds approximate what is fundamentally a continuous along-belt metallogenic gradient.
6. **Reporting scope of uncertainty:** Stored out-of-fold prospectivity predictions represent expected conditional probabilities $\bar{p}_i$; cell-level credible intervals and simulated Bernoulli outcome distributions ($Y_i^{(s)} \sim \text{Bernoulli}(p_i^{(s)})$) were not part of the frozen reporting tables.

7 Conclusions

The hierarchical Bayesian model represents posterior uncertainty in global and domain-varying parameters, and the frozen held-out scores are posterior mean conditional probabilities. The OOF

decomposition shows geographically varying SD-normalized shares of additive linear-predictor components, while fold and domain discrimination vary across the evaluated sectors. These are model-based associations and predictive results, not causal evidence that controls or ore-forming processes change between domains. Distance associations recur across the evaluated representations; quadratic curvature and derived D^* diagnostics are representation-dependent, and population-level D^* is unidentifiable. Spatially separated validation provides an evaluation for the specified folds, not a guarantee of transfer to all frontier areas.

8 Reproducibility, data, and code availability

The modeling workflow is implemented in Python using PyMC (Salvatier et al., 2016), ArviZ, and scikit-learn. The core spatial cross-validation script is `src/v11_spatial_stability_final.py`; OOF extraction is in `src/extract_v11_oof.py`; the distance-sensitivity and population-reconstruction scripts are also identified in the repository. The frozen V11 sampler call uses two chains, 2,500 tuning steps, 1,500 retained draws per chain, target acceptance 0.99, one core, and random seed 42. The original input CSV, four frozen traces, and OOF means are in the repository. The original Bouguer text grid is referenced by a drive-local path in an executed notebook and is not archived here; complete original-source metadata for contact linework and occurrence records is also unavailable in the files examined. These provenance gaps remain limitations for independent reproduction.

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